Dark Energy from Dark-Dimension Modes with a Running Ansatz

Mikal J. Nolan

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Abstract

The cosmological-constant problem, a discrepancy of up to $\sim 10^{120}$ between naive quantum-field-theory vacuum-energy estimates and the observed dark-energy density, remains unresolved. Recent DESI baryon-acoustic-oscillation analyses, especially when combined with CMB and supernova data, show dataset-dependent hints that the dark-energy sector may be better described by an evolving equation of state than by a strict cosmological constant. This paper presents a deliberately narrow phenomenological framework in which the observed dark-energy scale is tied to a mesoscopic dark dimension. We adopt the dark-dimension scaling relation $\rho_{\mathrm{DE}} \propto R_{\mathrm{DD}}^{-p}$, motivated by the swampland dark-dimension scenario, rather than deriving the vacuum energy from an explicit Kaluza-Klein stress-energy calculation. The absolute scale of the dark-dimension radius today is fixed only after specifying the matching coefficient in this scaling. The new content of the framework is a separate postulate that the running of the dark-dimension radius tracks a slowly evolving parent-horizon scale. We work through the resulting effective dark-energy equation of state in the quasi-de Sitter regime and identify a specific correlation between the inferred dark-energy equation-of-state parameter and the cosmological running of Newton's constant via the five-dimensional Planck relation. The correlation is independent of the detailed running ansatz, but only within the simplest unscreened realization in which the same modulus controls both ρ_{DE} and the measured four-dimensional Newton constant. Limitations are stated explicitly: a first-principles bulk-flow computation of ρ_{DE}, a derivation of the running postulate from higher-dimensional dynamics, and a self-consistent scalar-tensor cosmological treatment are left for future work.

1 Introduction

1.1 The cosmological-constant problem and the observed acceleration

The cosmological-constant problem is one of the most severe fine-tuning issues in physics. Naive quantum-field-theory estimates of vacuum energy vastly exceed the observed dark-energy density responsible for cosmic acceleration. Recent DESI baryon-acoustic-oscillation results, when combined with CMB and supernova data, show dataset-dependent evidence that a time-dependent dark-energy equation of state may fit the data better than a strict cosmological constant.

1.2 Strategy of this paper

Let us adopt a deliberately narrow and falsifiable strategy:

1. Adopt an FLRW cosmology as the background spacetime for the dark-dimension running ansatz.

2. Use the dark-dimension scenario as a phenomenological explanation for the mesoscopic dark-energy scale.

3. Postulate, as the substantive content of the framework, that the running of the dark-dimension radius with cosmic scale factor is set by a parent-horizon scale, while the absolute normalization today is fixed by matching to the observed dark-energy density after specifying the coefficient in the scaling relation.

4. Translate the running postulate into an effective dark-energy equation of state.

5. Identify a cross-prediction relating the inferred dark-energy equation-of-state parameter to the running of Newton's constant via the five-dimensional Planck relation, and show that this cross-prediction is independent of the specific functional form of the running.

I do not adopt the upward extension of the cosmological-coupling proposal. The framework presented below requires only a slowly evolving parent geometry, which may be produced by accretion or by another slow higher-dimensional process; no specific interior metric is required.

2 Theoretical Foundations

2.1 The dark dimension

The dark-dimension proposal posits a single compact extra dimension of mesoscopic size. In this paper I adopt the associated phenomenological scaling

$$\rho_{\mathrm{DE}} \simeq \frac{\alpha}{R_{\mathrm{DD}}^{p}}, \qquad p \simeq 4. \tag{1}$$

Here α is a matching coefficient whose mass dimension depends on p. For $p = 4$ in natural units, α may be treated as dimensionless; for other values of p, it carries the corresponding compensating dimension. The present radius is therefore

$$R_{\mathrm{DD},0} = \left(\frac{\alpha}{\rho_{\mathrm{DE},0}}\right)^{1/p}. \tag{2}$$

Thus the observed dark-energy density fixes $R_{\mathrm{DD},0}$ only after α is specified or estimated from the underlying theory. The power $p = 4$ corresponds to the optimistic end of the swampland-motivated dark-dimension scaling, and I adopt it throughout unless otherwise stated. The matching coefficient absorbs differences in p for the present-day absolute scale, but the cross-prediction slope derived below depends linearly on p; this dependence is reinstated explicitly where relevant. The framework is constrained by tabletop tests of the inverse-square law.

2.2 Higher-dimensional context

The framework sits within the broader landscape of higher-dimensional gravity, in which IR modifications to gravitational dynamics can arise from extra-dimensional physics, for example in DGP-type braneworlds. I do not import any specific such mechanism; I use only the dark-dimension parametric scaling (1) together with the running postulate stated below.

3 Phenomenological Model and Toy Calculation

3.1 Cutoff identification: absolute scale and the running postulate

Absolute scale. As in the original dark-dimension scenario, the absolute scale of the dark-dimension radius today is fixed by demanding that the vacuum-energy formula (1) reproduce the observed dark-energy density, once the coefficient α has been specified. The consistency condition involves only the present dark-energy density and the chosen dark-dimension scaling; the parent-horizon scale does not set the present-day normalization.

Running postulate. The substantive content of the framework is a separate postulate: the time evolution of the dark-dimension radius is locked to a slowly evolving parent-horizon scale through

$$R_{\mathrm{DD}}(t) = R_{\mathrm{DD},0}\left(\frac{R_{\mathrm{parent}}(t)}{R_{\mathrm{parent},0}}\right)^{n}, \tag{3}$$

where the exponent n parametrizes the ansatz. The original $\sqrt{\cdot}$ form corresponds to $n = 1/2$. Differentiating gives

$$\frac{d\ln R_{\mathrm{DD}}}{d\ln a} = n\frac{d\ln R_{\mathrm{parent}}}{d\ln a}. \tag{4}$$

The parent quantity in (3) should be understood as a horizon-radius scale, not necessarily as an ADM mass. In a D-dimensional Schwarzschild-Tangherlini geometry, an ADM mass scales as $M_{\mathrm{ADM}} \propto R_h^{D-3}$, so identifying the running variable directly with a physical mass would change the exponent. To avoid this ambiguity, the model below uses the dimensionless parent-horizon variable

$$Q(t) \equiv \frac{R_{\mathrm{parent}}(t)}{R_{\mathrm{parent},0}}. \tag{5}$$

For the dynamical predictions below, only the combination $n\,d\ln Q/d\ln a = d\ln R_{\mathrm{DD}}/d\ln a$ enters. Any mechanism producing the same evolution of R_{DD} would yield the same leading predictions.

3.2 Effective equation of state

Define the dimensionless elapsed time from the present epoch by

$$x \equiv H_0(t - t_0), \qquad x = 0 \quad \text{today}. \tag{6}$$

In an exactly de-Sitter background, $x = \ln(a/a_0)$. In the late universe this is only an approximation; more generally, $d\ln a/dx = H(a)/H_0$. The quasi-de Sitter toy model used here takes $H(a) \simeq H_0$ over the limited interval of interest.

Adopt the linear parent-horizon ansatz

$$Q(x) = 1 + \beta x, \qquad |\beta x| \ll 1. \tag{7}$$

Then

$$\frac{d\ln Q}{d\ln a} \simeq \frac{\beta}{1+\beta x}, \tag{8}$$

where the equality is exact in de Sitter and the displayed form receives a factor $H_0/H(a)$ in a general FLRW background. At the present epoch, $x = 0$, this gives $(d\ln Q/d\ln a)_0 \simeq \beta$.

Since $\rho_{\mathrm{DE}} \propto R_{\mathrm{DD}}^{-p} \propto Q^{-pn}$,

$$\frac{d\ln \rho_{\mathrm{DE}}}{d\ln a} = -pn\frac{d\ln Q}{d\ln a}. \tag{9}$$

Treating dark energy as an effective fluid, an operational definition discussed below, the continuity equation gives

$$1 + w_{\text{eff}} = -\frac{1}{3}\frac{d\ln\rho_{\text{DE}}}{d\ln a} = \frac{pn}{3}\frac{d\ln Q}{d\ln a}. \tag{10}$$

For the linear ansatz in the quasi-de Sitter approximation,

$$1 + w_{\text{eff}}(x) \simeq \frac{pn}{3}\frac{\beta}{1+\beta x}. \tag{11}$$

Thus at the present epoch

$$w_0 = -1 + \frac{pn\beta}{3} = -1 + \frac{4n\beta}{3} \quad (p = 4). \tag{12}$$

The statement that linear parent-horizon growth produces an exactly constant w is therefore limited to the leading small-βx, quasi-de Sitter approximation. In a full FLRW treatment, even constant $\dot{Q}$ generally produces mild scale-factor dependence through $H(a)$.

For the quadratic extension

$$Q(x) = 1 + \beta x + \gamma x^2, \qquad |\beta x|,\ |\gamma x^2| \ll 1, \tag{13}$$

one finds

$$\frac{d\ln Q}{d\ln a} \simeq \frac{\beta + 2\gamma x}{1 + \beta x + \gamma x^2} \simeq \beta + 2\gamma x \tag{14}$$

in the same quasi-de Sitter and slow-running regime. Therefore

$$1 + w_{\text{eff}}(a) \simeq \frac{pn}{3}\left[\beta + 2\gamma\ln(a/a_0)\right]. \tag{15}$$

Using the CPL form $w(a) = w_0 + w_a(1-a)$ and $\ln a = -(1-a) + O[(1-a)^2]$ near $a_0 = 1$, this gives

$$w_0 = -1 + \frac{pn\beta}{3}, \qquad w_a = -\frac{2pn\gamma}{3}. \tag{16}$$

For $p = 4$,

$$w_0 = -1 + \frac{4n\beta}{3}, \qquad w_a = -\frac{8n\gamma}{3}. \tag{17}$$

This demonstrates how the same logic produces both w_0 and w_a within the quadratic toy ansatz. It is not an ansatz-independent relation between w_0 and w_a.

Remark on the continuity equation. Because G runs in this framework, the energy-momentum balance is strictly that of a scalar-tensor theory, in which the modulus kinetic term and frame choice enter the Friedmann equations. The expression

$$1 + w_{\text{eff}} = -\frac{1}{3}\frac{d\ln\rho_{\text{DE}}}{d\ln a} \tag{18}$$

is therefore to be understood as the effective equation of state inferred by an observer fitting the dark-energy density to a power-law evolution at fixed G. For the slow-running regime $|d\ln G/d\ln a| \ll 1$, this matches the leading behavior. A fully self-consistent scalar-tensor analysis is left for future work.

3.3 Cross-prediction with the running of Newton's constant

For a simple circle compactification, the reduced Planck relation is, up to convention-dependent geometric factors,

$$M_{\rm Pl}^2 \propto M_5^3 R_{\rm DD}. \tag{19}$$

Holding the fundamental scale M_5 fixed gives

$$G \propto R_{\rm DD}^{-1}. \tag{20}$$

Therefore

$$\frac{d\ln G}{d\ln a} = -\frac{d\ln R_{\rm DD}}{d\ln a} = -n\frac{d\ln Q}{d\ln a}. \tag{21}$$

Combining (10) and (21), the factor $n\, d\ln Q/d\ln a$ cancels:

$$1 + w_{\rm eff} \simeq -\frac{p}{3}\frac{d\ln G}{d\ln a}. \tag{22}$$

At the present epoch,

$$1 + w_0 \simeq -\frac{p}{3}\left.\frac{d\ln G}{d\ln a}\right|_0 = -\frac{4}{3}\left.\frac{d\ln G}{d\ln a}\right|_0 \qquad (p=4). \tag{23}$$

This is a straight line through the origin with slope $-p/3 = -4/3$ when plotting $1 + w_0$ against $d\ln G/d\ln a$, provided the same cosmological modulus controls both $\rho_{\rm DE}$ and the measured Newton constant.

Why the result is independent of n. Both $1 + w_{\rm eff}$ and $d\ln G/d\ln a$ depend on the running ansatz only through $d\ln R_{\rm DD}/d\ln a$, since $\rho_{\rm DE}$ and G are fixed power-law functions of $R_{\rm DD}$ alone in the simplest model. The running rate cancels in the ratio. The cross-prediction therefore tests the joint assumptions

$$\rho_{\rm DE} \propto R_{\rm DD}^{-p}, \qquad G \propto R_{\rm DD}^{-1}, \qquad M_5 = \text{constant}, \tag{24}$$

plus the absence of screening, localization, or frame effects that decouple local measurements of G from the cosmological modulus. A violation of (23) would rule out this simplest unscreened realization. It would not, by itself, rule out every possible dark-dimension model.

4 Observational Consistency and a Distinctive Cross-Prediction

4.1 Background-level consistency

In the quasi-de Sitter and slow-running limit, the model reduces to an approximately constant-w dark-energy cosmology for linear parent-horizon growth and to ΛCDM in the limit of vanishing growth rate. For moderate positive β, the predicted sign $w_0 > -1$ is consistent with the quadrant favored by some DESI+CMB and DESI+CMB+supernova combinations in w_0w_aCDM fits. This should be read only as a background-level diagnostic, since those observational constraints are usually inferred within fixed-G GR parameterizations, while the present framework contains a running modulus.

4.2 A distinctive cross-prediction and an existing tension

The framework predicts a definite line in the plane of inferred dark-energy equation-of-state parameter versus cosmological running of Newton's constant. For an illustrative deviation

$$1 + w_0 \sim 0.05, \tag{1}$$

the cross-prediction implies

$$\left|\frac{\dot{G}}{G}\right| \sim \frac{3}{4}(1 + w_0)H_0 \sim 2.7 \times 10^{-12}\,\mathrm{yr}^{-1}, \tag{25}$$

where $H_0 \simeq 7.2 \times 10^{-11}\,\mathrm{yr}^{-1}$ has been used. This is more than an order of magnitude above the lunar-laser-ranging scale $\sim 10^{-13}\,\mathrm{yr}^{-1}$. I do not invoke an unmotivated screening mechanism to alleviate this tension. The simplest version of the relation survives only if locally measured Newton's constant is somehow decoupled from the cosmologically evolving bulk modulus, or if additional scalar-tensor, brane-localized, or frame effects modify the naive relation. As things stand, the cross-prediction is a sharp and currently tensioned prediction.

4.3 Auxiliary predictions

KK graviton modes may contribute to a stochastic gravitational-wave background, depending on the detailed production and leakage mechanism. This should be regarded as a possible inherited signature of dark-dimension phenomenology rather than as a quantitative prediction of the present toy model. If the higher-dimensional bulk source carries net angular momentum, a small large-scale anisotropy could in principle be imprinted on the cosmology, though no quantitative mechanism is derived here.

4.4 Falsifiability

The simplest unscreened realization is falsified if future data converge on a $(1+w_0, d\ln G/d\ln a)$ pair inconsistent with the predicted slope after accounting for the frame and local-versus-cosmological measurement issue. The broader framework is also constrained, and potentially excluded, if submillimetre gravity tests rule out the required dark-dimension window. Within a specified nonlinear parent-horizon ansatz, the corresponding predicted relation among w_0, w_a, and the running parameters can also be tested, but there is no ansatz-independent w_0-w_a relation.

5 Final Thoughts

Trading the enormous vacuum-energy fine tuning for a single mesoscopic geometric scale is a feature of the dark-dimension scenario itself. What the present framework adds is a phenomenological postulate tying the running of the dark-dimension radius to a parent-horizon scale, thereby providing a possible physical origin for time variation in the effective dark-energy density. The resulting cross-prediction between the inferred dark-energy equation of state and the running of Newton's constant is robust against changes in the detailed running ansatz, because the running rate cancels in the ratio. This robustness strengthens the prediction within the simplest unscreened model, but it also makes the current tension with bounds on $\dot{G}/G$ especially important.

Although the focus is late-time dark energy, the synthesis admits one qualitative early-universe imprint worth investigating in future work: if the running postulate holds throughout cosmic history, then the effective KK threshold was correspondingly different in the early universe. A consistent treatment of this statement would require extending the toy model beyond the quasi-de Sitter approximation.

6 Limitations

The framework is phenomenological. The principal omissions are a derivation of the running postulate from explicit higher-dimensional bulk dynamics, a derivation of the dark-energy density from explicit bulk stress-energy flow, a first-principles computation of the accretion or horizon-growth parameters, and a quantitative model of the parent-horizon evolution. A self-consistent scalar-tensor treatment of the cosmological dynamics with running G is also outstanding. Two further substantive limitations should be flagged. First, the requirement that the parent-horizon scale evolve slowly enough to match the inferred dark-energy dynamics relocates part of the original fine-tuning question to a model parameter. Second, the cross-prediction is at present in significant tension with existing bounds unless local and cosmological moduli decouple, or unless the naive relation between R_{DD} and locally measured G is modified.

Open problems include:

- Derivation of the running postulate from explicit higher-dimensional bulk dynamics, including a stabilization mechanism.
- Quantitative model of the slowly evolving bulk source or parent-horizon scale.
- Self-consistent scalar-tensor analysis of the cosmological dynamics with running G.
- Explicit computation of the dark-energy density from KK modes, bulk stress-energy, or another first-principles mechanism.
- Quantitative estimate of possible stochastic gravitational-wave signatures.
- Forecast of the constraining power of future surveys plus improved bounds on $\dot{G}/G$.
- Investigation of mechanisms that could decouple laboratory measurements from the cosmologically evolving bulk modulus.

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